

DUY-TUNG PHAM

+84 942806486 ◊ Hanoi, Vietnam ◊ [Email](#) ◊ [LinkedIn](#) ◊ [Github](#) ◊ [Google Scholar](#) ◊ [Homepage](#)

EDUCATION

Hanoi University of Science and Technology (HUST)

2020 - 2024

Bachelor of Science in Computer Science (Honors Program), CPA: 3.91/4.00, ranked 2nd in program.

Thesis: Fine-grained Neural Topic Modeling, resulting in a [paper](#) accepted to Findings of EMNLP 2024.

RESEARCH EXPERIENCE

AI Research Resident

Jun 2024 - Present

FPT Software AI Center

Hanoi, Vietnam

- Advisors: [Prof. Thieu Vo](#), [Prof. Tan Nguyen](#)
- Research topic: Properties of Foundation Model (Transformer, Mixture of Experts, Mamba), Optimal Transport.
- 4 accepted papers at ICLR, ICML, and NeurIPS.

Undergraduate Research Student

Dec 2021 - Jul 2024

DSLAb, HUST

Hanoi, Vietnam

- Advisors: [Dr. Linh Ngo Van](#), [Prof. Khoat Than](#)
- Research topic: Variational Inference, Bayesian Neural Network, Topic Modeling.
- 3 accepted papers at EMNLP and Neurocomputing journal.

PUBLICATIONS

- [Mixed-Curvature Tree-Sliced Wasserstein Distance](#)
Duy-Tung Pham*, Viet-Hoang Tran*, Thieu Vo, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR, rank A*), 2026.
- [Dynamical Properties of Tokens in Self-Attention and Effects of Positional Encoding](#)
Duy-Tung Pham*, An Nguyen The*, Viet-Hoang Tran, Nhan-Phu Chung, Xin T. Tong, Tan Minh Nguyen**, Thieu Vo**.
Conference on Neural Information Processing Systems (NeurIPS, rank A*), 2025.
- [TopiCOT: Neural Topic Model Aligning with Pre-trained Clustering and Optimal Transport](#)
Tung Nguyen*, **Duy-Tung Pham***, Quang Duc Nguyen, Linh Ngo Van, Duc Anh Nguyen, Sang Dinh Viet.
Neurocomputing (Q1 journal, IF 5.5), 2025.
- [Equivariant Polynomial Functional Networks](#)
Thieu Vo*, Viet-Hoang Tran*, Tho Tran Huu, An Nguyen The, Thanh Tran, Minh-Khoi Nguyen-Nhat, **Duy-Tung Pham**, Tan Minh Nguyen.
International Conference on Machine Learning (ICML, rank A*), 2025.
- [Demystifying the Token Dynamics of Deep Selective State Space Models](#)
Thieu Vo, **Duy-Tung Pham**, Xin T. Tong**, Tan Minh Nguyen**.
International Conference on Learning Representations (ICLR, rank A*, **spotlight**), 2025.
- [Equivariant Neural Functional Networks for Transformers](#)
Viet-Hoang Tran*, Thieu Vo*, An Nguyen The*, Tho Tran Huu, Minh-Khoi Nguyen-Nhat, Thanh Tran, **Duy-Tung Pham**, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR, rank A*), 2025.
- [Out-of-Vocabulary Handling and Topic Quality Control Strategies in Streaming Topic Models](#)
Tung Nguyen*, **Duy-Tung Pham***, Linh Ngo Van, Ha-Bang Ban, Khoat Than.
Neurocomputing (Q1 journal, IF 5.5), 2025.
- [NeuroMax: Enhancing Neural Topic Modeling via Maximizing Mutual Information and Group Topic Regularization](#)
Duy-Tung Pham*, Trang Nguyen*, Tung Nguyen*, Linh Ngo Van, Duc Anh Nguyen, Thien Huu Nguyen.
Findings of The Conference on Empirical Methods in Natural Language Processing (EMNLP, rank A*), 2024.

(*) denotes equal contribution, (**) denotes equal advising.

RESEARCH INTEREST

I study the *mathematical foundations of deep learning*, focusing on *geometric* and *dynamical* perspectives to design *efficient*, *scalable*, and *robust* models from first principles.

PROFESSIONAL SERVICES

Reviewer at ICLR 2025, ICLR 2026.

SCHOLARSHIPS

- **Academic Achievement Scholarship - HUST** (6/7 semesters): Awarded to the top 3% of students with outstanding academic performance each semester.
- **Scholarship from the National Key Program for Mathematical Development** (2018): Awarded by the Vietnam Institute for Advanced Studies in Mathematics (VIASM) to high school students demonstrating exceptional excellence and strong potential in mathematics.

RELEVANT COURSES AND CERTIFICATES

- **University Courses:** Calculus 1, 2, 3; Algebra; Optimization Methods; Discrete Mathematics; Probability and Statistics; Introduction to AI; Machine Learning (all achieved 4.0/4.0).
- **Language:** English (IELTS 7.5 Overall - Listening 8.5, Reading 8.5, Writing 6.5, Speaking 6.0), Vietnamese (Native).

PRESENTATIONS

- *Demystifying the Token Dynamics of Deep Selective State Space Models*. ICLR 2025 Spotlight. Singapore.
- *Bayesian Neural Network*. Weekly seminar, DSLab, HUST. Hanoi, Vietnam, 2024.

HONORS AND AWARDS

- **SoICT Hackathon**, Routing Optimization Track (Oct 2023) - Finalist, Top 7.
- **ACM-ICPC**, Asia Hanoi Regional Contest (Mar 2022) - Third Prize.
- **Vietnam Olympiad of Informatics** for Undergraduate Students (Dec 2020) - Second Prize.
- **Hanoi Math Modeling Online Contest** (Jul 2020) - First Prize.
- **Vietnam Mathematical Olympiad** for High School Students (Jan 2019) - Third Prize.

ACTIVITIES

- **Mentor, MaSSP, Applied Mathematics** Jul 2023 - Aug 2023
Guided 4 high school students in a financial math project on fraud detection, covering data handling, modeling, evaluation, and presentation.
- **Academic Assistant, Hanoi Math Modeling Competition** Jun 2021 - May 2022
Guided teams in real-world problem modeling using linear/integer programming and time series methods, assisting in formulation, analysis, and presentations.
- **Mentee, PiMA Research Summer Camp** Jul 2021 - Aug 2021
Researched and implemented Locally Linear Embedding (LLE) for image compression, exploring theory and variants.
- **Mentee, MaSSP, Data Science and Machine Learning** Jun 2021
Completed intensive coursework and applied skills analyzing COVID-19 patient data.
- **Participant, VIASM Mathematical Summer School** Jul 2019
Attended advanced problem-solving training for Mathematics Olympiad preparation.

REFERENCES

- **Prof. Tan Nguyen**, Assistant Professor (Presidential Young Professor), National University of Singapore.
Email: tanmn@nus.edu.sg. Homepage: <https://tanmnguyen89.github.io/>
- **Prof. Thieu Vo**, Lecturer (eq., Assistant Professor), University of Bath, United Kingdom.
Email: ntv22@bath.ac.uk. Homepage: <https://sites.google.com/tdtu.edu.vn/vongochthieu>
- **Prof. Xin. T. Tong**, Associate Professor, National University of Singapore.
Email: xin.t.tong@nus.edu.sg. Homepage: <https://sites.google.com/view/xintongthomson/home>
- **Dr. Linh Ngo Van**, Lecturer (eq., Assistant Professor), Hanoi University of Science and Technology, Vietnam.
Email: linhnv@soict.hust.edu.vn. Homepage: <https://users.soict.hust.edu.vn/linhnv/>